



POLITECNICO
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SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

GeoInformatics Project

Links to External Resources and User Manual

Webcam-Based PM2.5 Environmental Data Processing Framework

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1. Introduction

The purpose of this document is to provide the technical information required to access, reproduce, and operate the developed framework. It serves as a reference for the project repository, software organization, workflow implementation, and final dataset generation procedures.

This document is intended for researchers, students, and future users working with the framework or extending its functionality.

2. External Links

2.1 GitHub Repository

The complete project repository is available on GitHub:

GitHub Repository:

<https://github.com/QingxuanTuo/webcam-pm25-toolbox>

The repository contains the source code, Jupyter notebooks, configuration files, workflow modules, processed datasets, and technical documentation developed for the environmental data processing framework.

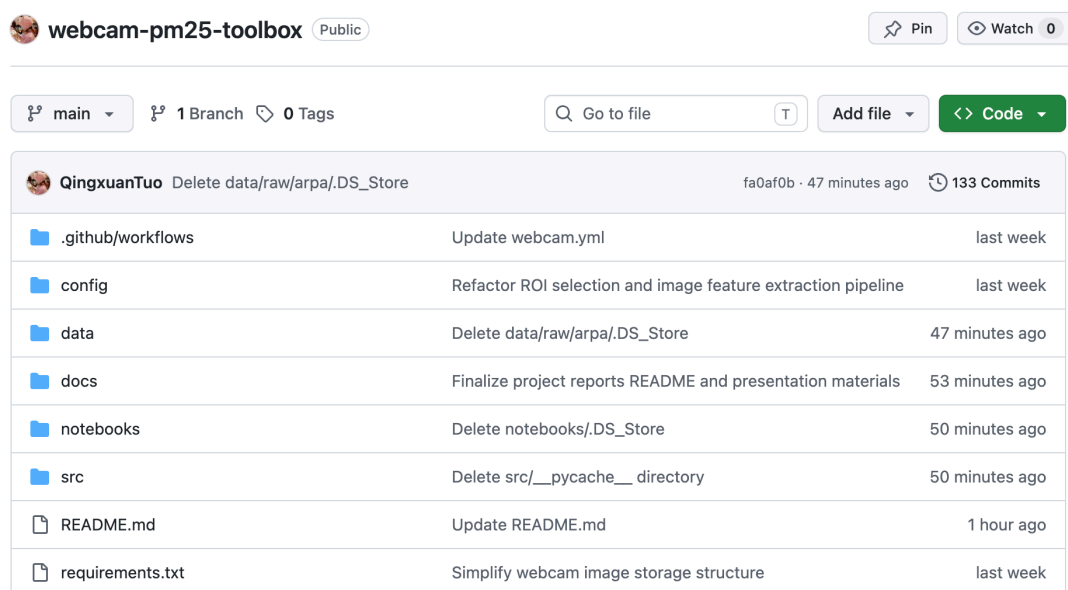


Figure 2.1 Main structure of the GitHub repository

Figure 2.1 illustrates the main structure of the project repository, including workflow notebooks, modular Python processing scripts, configuration files, and processed datasets.

The repository is organized as a modular and reproducible Python-based workflow designed to support future reuse, extension, and environmental data analysis applications.

2.2 External Data Sources

The framework integrates multiple external environmental data sources used for image collection, air quality monitoring, and meteorological data retrieval.

The following external platforms and environmental datasets were used throughout the project.

2.2.1 European Environment Agency (EEA) — PM2.5 Observations

Official Website:

<https://eeadmz1-downloads-webapp.azurewebsites.net>

The European Environment Agency (EEA) provides the PM2.5 air quality observations used in this project.

The framework retrieves hourly PM2.5 measurements corresponding to the monitoring station: IT/SPO.IT0477A_6001_BETA (MILANO - V.LE MARCHE)

The downloaded PM2.5 observations are subsequently standardized and integrated with webcam-derived image features and meteorological datasets during the environmental data processing workflow.

2.2.2 Copernicus Climate Data Store (CDS)

Official Website:

<https://cds.climate.copernicus.eu/>

The Copernicus Climate Data Store (CDS) provides the ERA5 meteorological reanalysis datasets used in the project.

The framework uses the following ERA5 datasets:

ERA5 Hourly Time-Series Data on Single Levels from 1940 to Present

This dataset provides hourly near-surface meteorological variables, including temperature, dew point temperature, wind components, surface pressure, precipitation, boundary layer height, and cloud-related variables.

Additional variables, including Relative Humidity (RH) and Wind Speed (WS10), were derived during preprocessing.

ERA5 Hourly Data on Pressure Levels from 1940 to Present

This dataset provides upper-air atmospheric variables, including geopotential, temperature, and wind components.

The project mainly uses the 500 hPa and 850 hPa pressure levels to represent large-scale atmospheric conditions related to PM2.5 transport and meteorological variability.

2.2.3 ARPA Lombardia

Official Website:

<https://www.arpalombardia.it/temi-ambientali/meteo-e-clima/form-richiesta-dati>

ARPA Lombardia provides local ground meteorological observations for the Lombardy region.

The project uses environmental observations corresponding to the MILANO - V.LE MARCHE monitoring station in order to maintain spatial consistency with the webcam imagery and PM2.5 observations used throughout the framework.

The integrated ARPA datasets include temperature, relative humidity, wind speed, wind direction, and gust wind measurements collected at hourly temporal resolution.

2.2.4 Webcam Data Source

Official Website:

<https://www.meteogiuliacci.it/meteo-webcam/webcam-milano>

Webcam imagery used in this project was collected from the MeteoGiuliacci webcam platform.

The webcam platform continuously updates webcam images throughout the day, while historical images are not permanently archived on the website. To preserve the temporal image records required for environmental analysis, the framework uses automated Python scripts to download webcam images at regular intervals.

The downloaded hourly webcam images are subsequently used for ROI analysis and image feature extraction.

3. System Requirements

The following system configuration is recommended for executing the environmental data processing framework and reproducing the workflow presented in this project.

Table 3.1 Summary of system requirements :

Requirement	Description
Operating Systems	Windows, Linux, macOS
Python Version	Python 3.10 or higher
Development Environment	Jupyter Notebook, VS Code, PyCharm, or Google Colab
Internet Connection	Required for external data downloading
External Accounts	Copernicus Climate Data Store (CDS) account
Storage	Sufficient local or cloud storage recommended

3.1 Python Environment

All required Python dependencies are provided in the requirements.txt file included in the project repository.

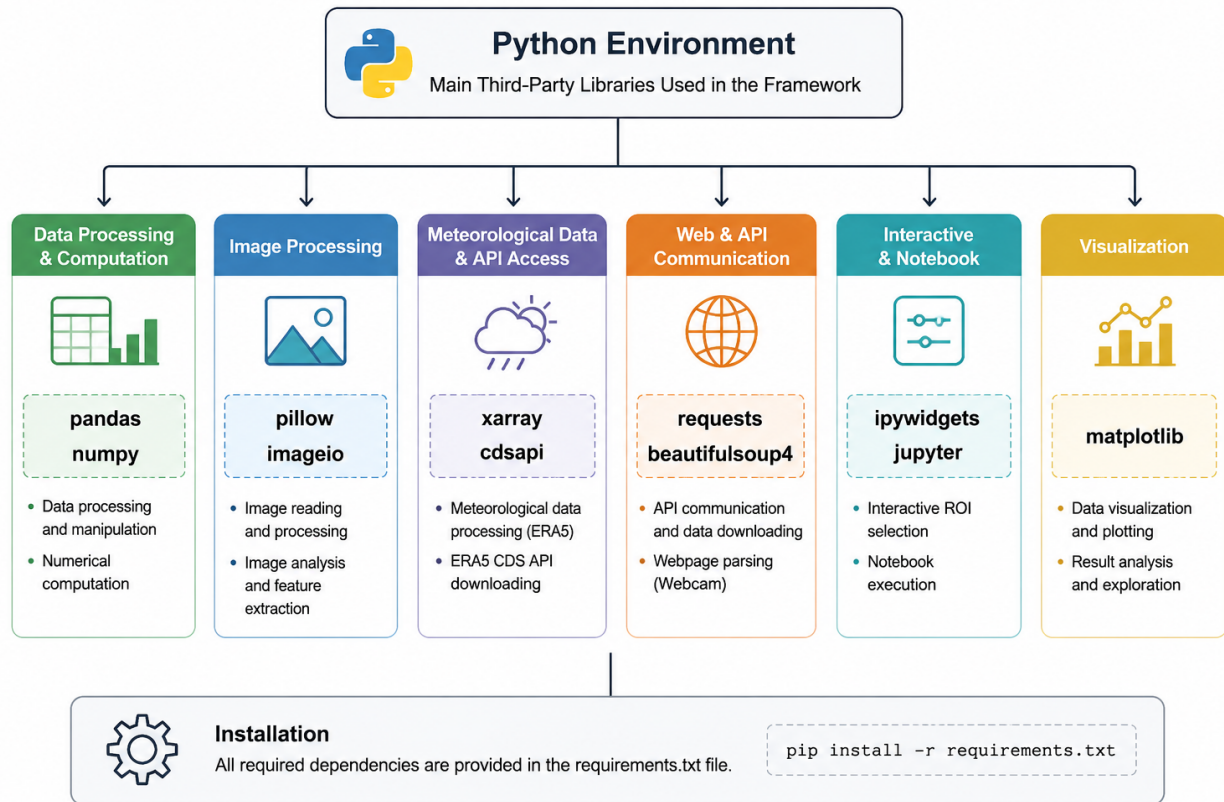


Figure 3.1 Main Python library categories used in the framework.

Install the required dependencies using: *pip install -r requirements.txt*

3.2 Development Environment

The main workflow is executed through Jupyter notebooks, while reusable Python modules are organized within the src/ directory.

3.3 Internet Connection Requirements

An active internet connection is required for downloading external environmental datasets and executing the automated data collection workflows implemented in the framework.

Main online operations include:

- Webcam image downloading
- EEA PM2.5 API access
- ERA5 CDS API access
- External environmental data retrieval

3.4 External Accounts

A Copernicus Climate Data Store (CDS) account is required for downloading ERA5 meteorological datasets.

Before executing the ERA5 downloading workflow, users must:

1. Register for a CDS account
2. Obtain the CDS API credentials
3. Configure the CDS API locally through the `.cdsapirc` configuration file

Detailed configuration instructions are provided in the project repository README documentation.

3.5 Storage Requirements

Sufficient local or cloud storage is recommended for long-term workflow execution, particularly for automated webcam image collection and environmental dataset archiving.

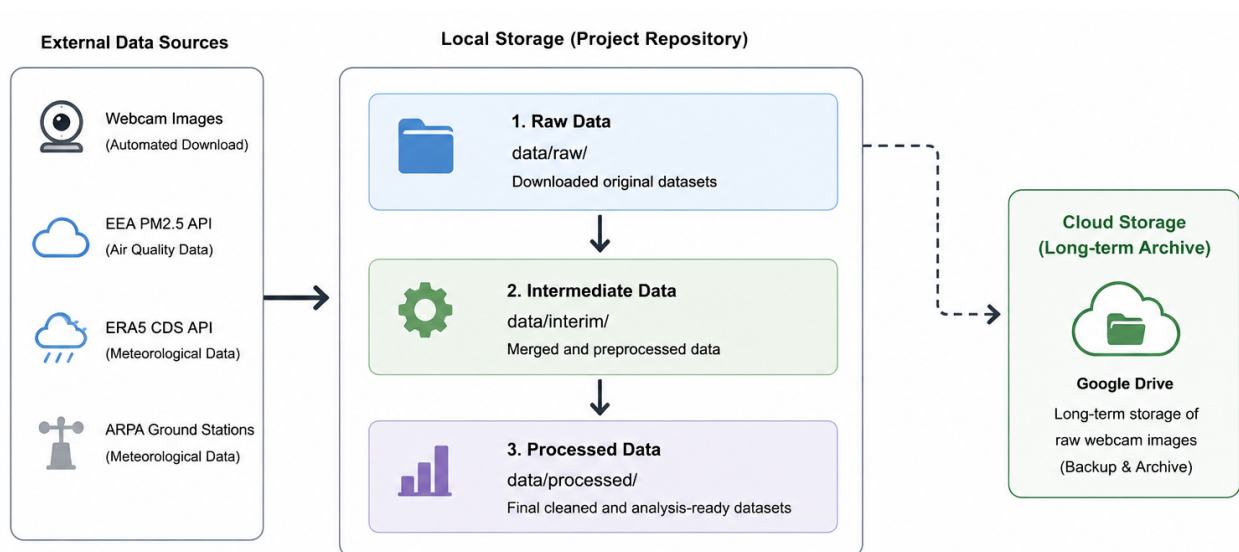


Figure 3.5 Storage architecture of the environmental data processing framework.

The framework primarily stores downloaded and processed datasets within the local data/ directory of the project repository, while long-term webcam image archives may additionally be preserved using external cloud storage services such as Google Drive.

4. Project Structure

The environmental data processing framework is organized as a modular Python project combining reusable processing scripts, Jupyter notebook workflows, configuration files, and structured environmental datasets.

webcam-pm25-toolbox/

```
|— .github/workflows
|   |— webcam.yml
|— config/
|   |— roi.json
|— data/
|   |— raw/
|       |— arpa/
|       |— eea_pm25/
|       |— era5/
|       |— webcam_images/
|   |— interim/
|   |— processed/
|— docs/
|— notebooks/
|   |— 01_webcam_pm25_dataset_pipeline.ipynb
|   |— 02_dataset_cleaning_preprocessing.ipynb
|— src/
|   |— __init__.py
|   |— webcam_download.py
|   |— roi_viewer.py
|   |— image_features.py
```

```
| |—— eea_pm25_download.py
| |—— era5_download.py
| |—— merge_arpa_data.py
| |—— merge_all_data.py
|—— requirements.txt
|—— README.md
```

5. Software Workflow

The main workflow of the project is executed through the following Jupyter notebooks:

- 01_webcam_pm25_dataset_pipeline.ipynb
- 02_dataset_cleaning_preprocessing.ipynb

The notebooks call reusable Python modules located in the `src/` directory to execute different environmental data processing tasks.

Step 1 — Automated Webcam Image Download

Required Files

- [src/webcam_download.py](#)
- [.github/workflows/webcam.yml](#)

User Setup

Before running the automated webcam downloading workflow, users must configure GitHub Actions and Google Drive authentication for automated image collection and cloud storage synchronization.

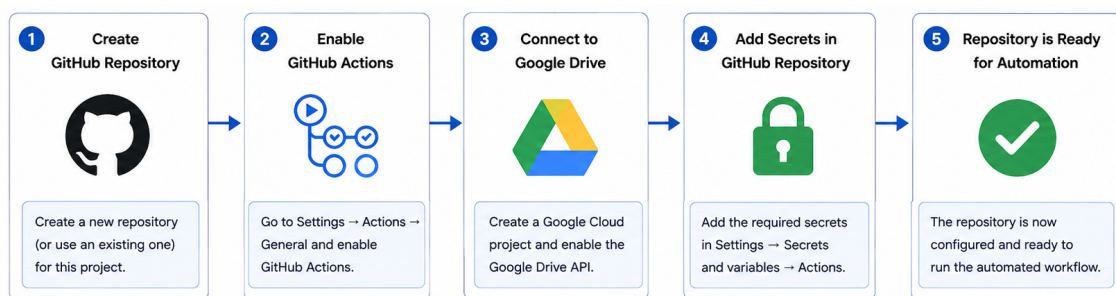


Figure 5.1 User Setup and Google Drive Configuration Workflow

Workflow Execution

The automated downloading workflow is executed through GitHub Actions, which periodically retrieves webcam images and uploads them to Google Drive for long-term storage.

Image Storage and Subsequent Processing

For ROI selection and image feature extraction workflows, two execution options are supported:

Option 1 — Google Colab

- Connect Google Colab to the same Google Drive account
- Directly access the webcam images

Option 2 — Local Execution

- Download webcam images from Google Drive
- Place images into: data/raw/webcam_images/

Step 2 — ROI Selection

Module: src/roi_viewer.py

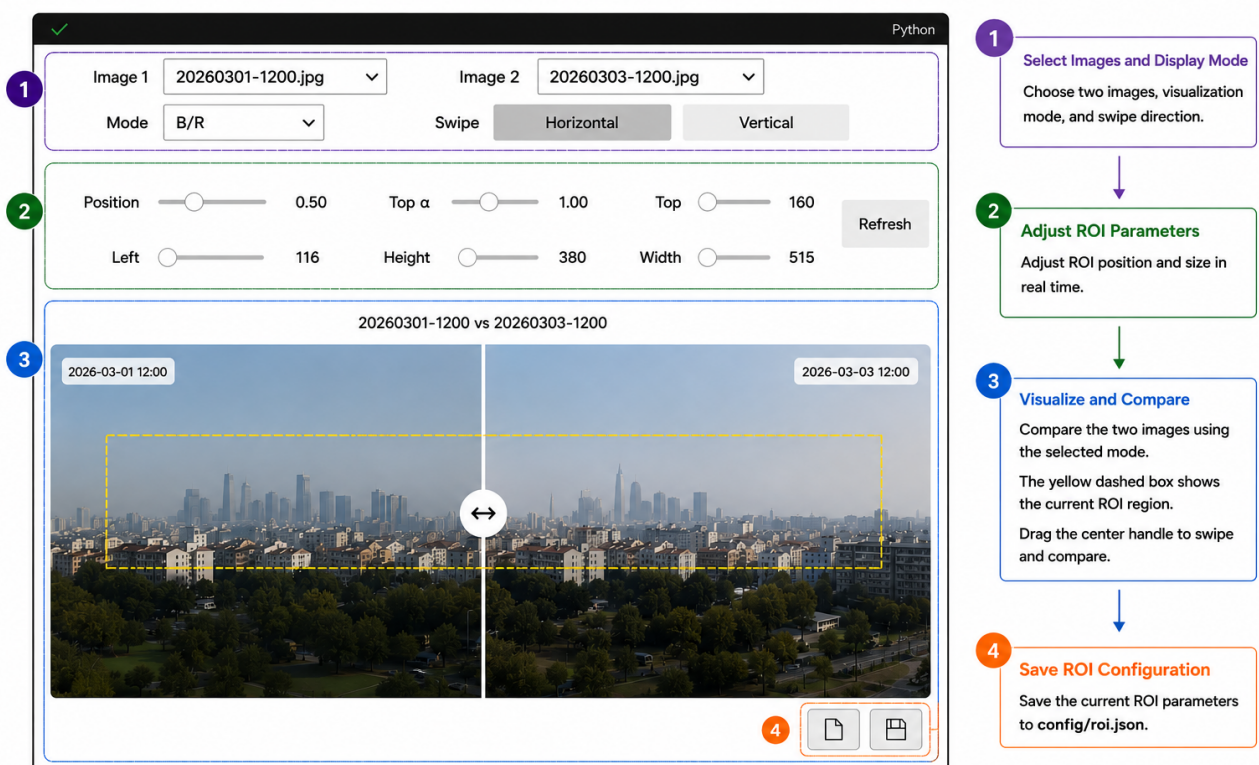


Figure 5.2 Interactive ROI Selection Workflow

The module provides an interactive interface for ROI selection, visualization, parameter adjustment, and ROI configuration saving for subsequent image feature extraction workflows.

Output: config/roi.json

Step 3 — Image Feature Extraction

Module: src/image_features.py

The module reads the ROI configuration from config/roi.json and extracts visual features from the selected ROI area of each webcam image.

Extracted features include:

- RGB statistics
- Saturation
- Blue-to-Red ratio
- Image contrast

Output: data/interim/image_features.csv

Step 4 — PM2.5 Data Download and Processing

Module: src/eea_pm25_download.py

Main Operations:

- Downloads EEA PM2.5 observations
- Parses parquet files
- Filters monitoring stations
- Converts timestamps to UTC
- Generates hourly PM2.5 datasets

Before execution, users may modify the target date range and monitoring station parameters in the script according to the required study period and monitoring location.

Output: data/interim/PM25_MI_hourly.csv

Step 5 — ERA5 Data Download and Processing

Module: [src/era5_download.py](#)

The module downloads ERA5 single-level and pressure-level meteorological datasets from the CDS API, processes the retrieved files, computes derived meteorological variables, and generates a unified hourly ERA5 dataset.

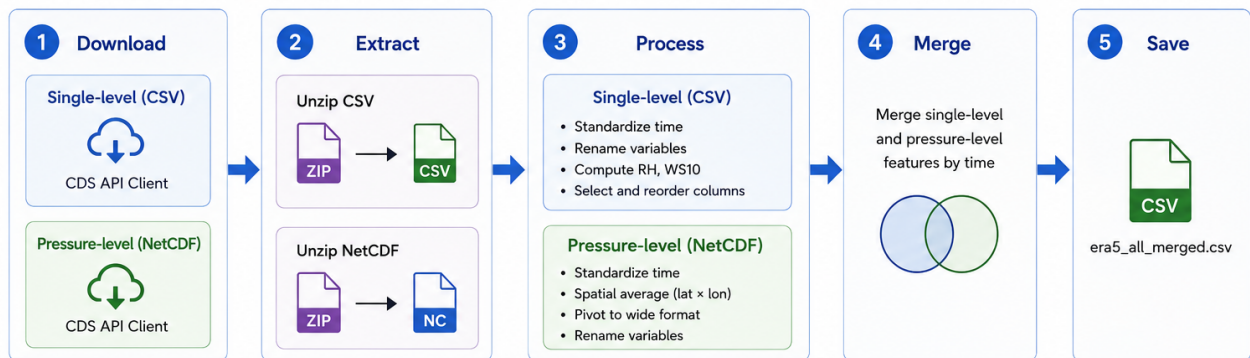


Figure 5.3 ERA5 Data Download and Processing Workflow

Output: [data/interim/era5_all_merged.csv](#)

Step 6 — ARPA Data Integration

Module: [src/merge_arpa_data.py](#)

Before executing the workflow, users must manually download the ARPA meteorological datasets from the ARPA Lombardia platform, extract the downloaded files, and place the CSV files into: `data/raw/arpa/`

The module subsequently merges and standardizes the ARPA meteorological observations into a unified hourly dataset.

Output: [data/interim/arpa_merged.csv](#)

Step 7 — Multi-source Dataset Integration

Module: [src/merge_all_data.py](#)

The module integrates webcam image features, PM2.5 observations, ERA5 meteorological variables, and ARPA ground meteorological datasets into a unified hourly dataset using UTC timestamps.

Output: [data/interim/merged_dataset.csv](#)

Step 8 — Data Cleaning and Feature Engineering

Notebook: [02_dataset_cleaning_preprocessing.ipynb](#)

The notebook loads the previously integrated dataset stored in the data/interim/ directory and performs dataset cleaning, preprocessing, validation, interpolation, temporal consistency checking, and feature engineering to generate the final analysis-ready dataset.

Generated Features:

- hour
- weekday
- month
- hour_sin
- hour_cos
- day/night indicator
- wind_dir_sin
- wind_dir_cos

Final output: [data/processed/final_dataset.csv](#)

6. Final Dataset

6.1 Dataset Overview

The final dataset generated by the framework integrates webcam image features, PM2.5 air quality observations, ERA5 meteorological variables, ARPA ground meteorological observations, and engineered temporal features into a unified hourly environmental dataset.

All variables are aligned using UTC hourly timestamps for subsequent environmental analysis and modeling workflows.

The final processed dataset is stored as: [data/processed/final_dataset.csv](#)

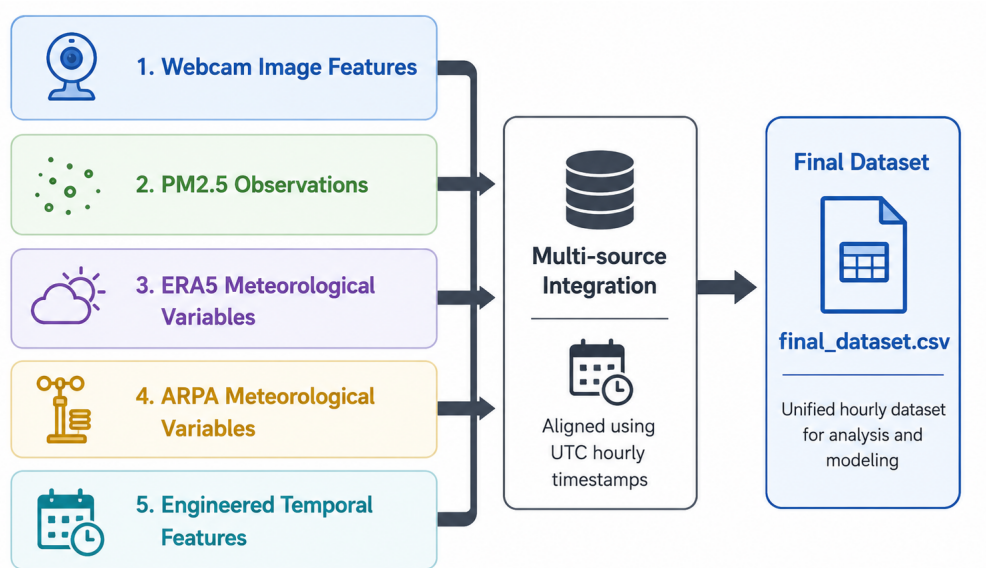


Figure 6.1 Final Dataset Structure Overview

6.2 Intended Applications

The generated dataset can support:

- PM2.5 prediction
- Air quality monitoring
- Environmental analysis
- Machine learning applications
- Urban environmental studies

7. Reproducibility

The environmental data processing framework follows a modular and reproducible workflow design.

All processing steps can be independently executed using the provided Python modules, Jupyter notebooks, configuration files, and workflow scripts included in the GitHub repository.

External environmental datasets are retrieved from publicly accessible platforms, while intermediate and final datasets are generated through standardized processing procedures.

The modular project structure additionally allows future users to modify, extend, or reproduce individual workflow components independently.

Annex — Final Dataset Schema

1. Dataset Overview

Item	Description
Dataset File	data/processed/final_dataset.csv
Temporal Resolution	Hourly UTC timestamps
Integrated Data Sources	Image features; PM2.5 observations; ERA5 meteorological variables; ARPA ground meteorological observations; engineered temporal features
Intended Applications	PM2.5 prediction; air quality analysis; environmental monitoring; machine learning applications; urban environmental studies

2. Dataset Characteristics

Property	Value
Dataset Name	final_dataset.csv
Temporal Resolution	Hourly
Time Zone	UTC

3. Variable Schema

3.1 Time Variables

variable	Description	Unit	Source
time	UTC timestamp	datetime	Generated
hour	Hour of day	-	Feature Engineering
weekday	Day of week	-	Feature Engineering
month	Month index	-	Feature Engineering
is_daytime	Day/night indicator	boolean	Feature Engineering

3.2 Webcam Image Features

Variable	Description	Source
R_roi	Mean red channel value inside ROI	Webcam
G_roi	Mean green channel value inside ROI	Webcam
B_roi	Mean blue channel value inside ROI	Webcam
S_mean	Mean saturation value	Webcam
B_R_ratio	Blue-to-Red ratio	Webcam
contrast	Image contrast	Webcam

3.3 PM2.5 Variables

Variable	Description	Unit	Source
PM25	PM2.5 concentration	$\mu\text{g}/\text{m}^3$	EEA

3.4 ERA5 Single-Level Meteorological Variables

Variable	Description	Unit	Source
T2M	2 m air temperature	°C	ERA5
D2M	2 m dew point temperature	°C	ERA5
RH	Relative humidity	%	Derived
U10	10 m zonal wind component	m/s	ERA5
V10	10 m meridional wind component	m/s	ERA5
WS10	Wind speed at 10 m	m/s	Derived
SP	Surface pressure	hPa	ERA5
TP	Total precipitation	mm	ERA5
BLH	Boundary layer height	m	ERA5

TCC	Total cloud cover	%	ERA5
CBH	Cloud base height	m	ERA5

3.5 ERA5 Pressure-Level Variables

Variable	Description	Unit	Source
GP_500	Geopotential at 500 hPa	gpm	ERA5
T_500	Temperature at 500 hPa	°C	ERA5
U_500	Zonal wind component at 500 hPa	m/s	ERA5
V_500	Meridional wind component at 500 hPa	m/s	ERA5
GP_850	Geopotential at 850 hPa	gpm	ERA5
T_850	Temperature at 850 hPa	°C	ERA5
U_850	Zonal wind component at 850 hPa	m/s	ERA5
V_850	Meridional wind component at 850 hPa	m/s	ERA5

3.6 ARPA Ground Meteorological Variables

Variable	Description	Unit	Source
temperature	Ground station temperature	°C	ARPA
wind_direction	Wind direction	degree	ARPA
relative_humidity	Ground station relative humidity	%	ARPA
wind_speed	Ground station wind speed	m/s	ARPA
wind_gust	Ground station gust wind	m/s	ARPA

3.7 Engineered Cyclical Features

Variable	Description	Source
hour_sin	Sine transformation of hour	Feature Engineering
hour_cos	Cosine transformation of hour	Feature Engineering
wind_dir_sin	Sine transformation of wind direction	Feature Engineering
wind_dir_cos	Cosine transformation of wind direction	Feature Engineering

4. Dataset Usage

The final dataset combines visual environmental information, atmospheric variables, air quality observations, and ground meteorological measurements into a unified hourly environmental dataset for:

- PM2.5 prediction models
- Environmental monitoring systems
- Urban air quality studies
- Computer vision applications for environmental sensing
- Machine learning and data mining research